Cross-Disaster Building Damage Assessment

Muhammad Murtaza and Darrin Leleux

Engineering Data Science, University of Houston

Abstract

Can a model trained on other disasters (earthquakes, wildfires, tornadoes) assess building damage in Houston after a flood, using few or no local labels?

Background

- **Need:** Rapid building damage assessment supports disaster response and recovery.
- **Challenge:** Models trained on other disasters may struggle with Houston flooding because damage patterns and local conditions differ.
- **Dataset:** xBD provides before-and-after satellite imagery with building outlines and damage labels.
- **Research focus:** Test transfer to Houston floods with no local training labels, then measure the benefit of limited local labeling.

Methods

Insert methods here

Insert figure here

Results

Insert results here

Conclusion

Insert conclusion here

Future Direction

Insert future direction here

Acknowledgments

Insert acknowledgements here